

Cooperative Relay-Assisted Traveling Salesman Problem using Graphs of Convex Sets

Satyanarayana G. Manyam¹, David W. Casbeer² and Isaac E. Weintraub²

Abstract—A cooperative routing problem involving two vehicles, a sensing agent and a relay agent, is considered. The sensing agent must visit a given set of targets to collect data and transmit it to a ground station. Hardware limitations restrict the transmission range, requiring the second agent to act as a relay between the sensing agent and the ground station. The collected data is transmitted through a two-hop network: from the sensing agent to the relay agent, and then to the ground station. Given a set of target locations, the routing problem aims to determine the optimal sequence of targets to be visited by the sensing agent, along with the corresponding positions of the relay agent, to minimize the combined travel cost of both vehicles. This problem is formulated as a mixed-integer convex program utilizing the framework of graphs of convex sets. The formulation avoids “big-M” constraints, resulting in a tighter convex relaxation. Furthermore, this approach is shown to be applicable to the neighborhood traveling salesman problem and can be extended to a three-agent scenario with two relay agents. Computational experiments are conducted to evaluate the proposed formulations and validate the tightness of the lower bounds obtained via convex relaxation.

I. INTRODUCTION

The Traveling Salesman Problem (TSP) [1] is a fundamental combinatorial problem that arises in several surveillance and reconnaissance applications [2], [3]. Given a set of cities and the cost of travel between every pair of cities, the TSP aims to find a minimum-cost tour that originates from the start location, referred to as the depot, visits every city exactly once, and returns to the depot. The TSP is often adapted to suit various applications; consequently, several variants, each with different additional constraints, have been formulated and solved in the literature.

Unmanned Aerial Systems (UAS) are widely used in monitoring and surveillance applications, and therefore, solving the TSP for such systems naturally introduces new constraints. In this paper, we consider a variant of the TSP with communication constraints. Specifically, we consider a surveillance application where small aircraft must visit a given set of target locations and gather data using on-board sensors. Due to the criticality of the application, the data must be relayed to a ground station as soon as it is collected. However, it is established that the available communication bandwidth is inversely proportional to the

transmission distance [4]. This limits the maximum distance from the ground station at which a target can be sensed by a UAS. In the proposed application, two UAS are considered: one UAS collects data from the targets, while the second acts as a relay between the sensing UAS and the ground station. This setup allows the sensing UAS to collect data from targets located beyond the distance limit imposed when using a single UAS. However, to ensure successful data transfer, both the distance between the two UAS and the distance between the relay UAS and the ground station must be less than a specified threshold distance. We refer to this threshold distance, which guarantees the necessary available bandwidth, as the communication radius.

An illustration of this setup with two UAS and two-hop communication links is shown in Fig. 1. In general, the communication distance may depend on the directionality of the antennas, obstacles present in the environment, and other factors. For simplicity, we assume that this communication radius is constant in all directions. The feasible communication radius between the sensing UAS and the relay UAS could be different from the communication radius between the relay UAS and the ground station.

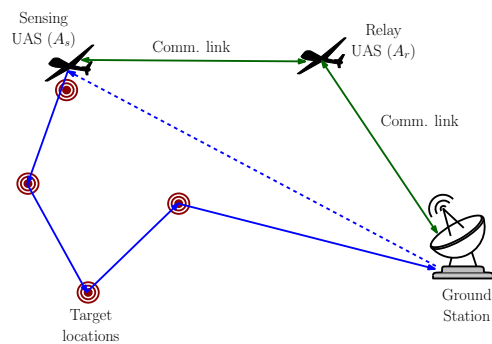


Fig. 1. Communication through a two-hop relay network, the blue lines represent the path plan while the green lines represent communication between systems.

The objective of this paper is to find a minimum-cost tour for the two UAS that starts from the ground station (the depot), visits every target, and returns to the depot, all while maintaining a feasible communication link from the sensing UAS to the ground station. We refer to this problem as the Communication Relay-Assisted TSP (CRATSP). An important distinction to observe is that while the route of the sensing UAS is defined over a discrete set of target locations, no such discrete set exists for the relay UAS. When the sensing UAS visits a target, the position of the relay UAS

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is a continuous variable restricted to a closed convex set. Therefore, the decision variables for this problem include the sequence of targets to be visited by the sensing UAS and the continuous positions of the relay UAS corresponding to each target. Note that, we use the terms “UAS” and “agent” interchangeably throughout this paper.

Literature Review: The Traveling Salesman Problem has been extensively studied [1], and some of its early variants are presented in [5]. Notable variants include the prize-collecting TSP [6], the TSP with time window constraints [7], the TSP with precedence constraints [8], the TSP with neighborhoods [9], [10], the generalized TSP [11], and the TSP with curvature constraints [12]–[14]. Although several variants of the TSP exist in the literature, the TSP for vehicles with communication and distance constraints remains relatively underexplored.

The variants that are closest to the proposed CRATSP are the ground vehicle–air vehicle coordinated TSP [15]–[18]. This class of problems shares a common theme: a ground vehicle carrying a drone travels on a road network, and the drone can take off at any target location to visit one or more targets before rejoining the ground vehicle. A key distinction, however, is that the drone can only take off and land at a pre-specified discrete set of locations. The variant considered in [18] aims to find a route for the ground vehicle defined as a series of connected line segments. The endpoints of these line segments are continuous decision variables. While this problem closely resembles the CRATSP, it does not incorporate communication constraints. Several other variants of the air-ground vehicle routing problem have been proposed and studied; interested readers may refer to the review papers [19], [20] for comprehensive overview. In our prior work on air-ground vehicle routing [21], communication constraints were enforced between the air and ground vehicles, but no such constraints existed between the vehicles and the ground station. Moreover, the vehicle positions were restricted to a discrete set, making the approach inapplicable to the CRATSP. A critical distinction of the CRATSP is that the position of the relay UAS is modeled as a continuous variable, an aspect not addressed in existing problem variants. This continuous variable formulation, coupled with the two-hop communication constraints, makes the CRATSP a novel problem and worthwhile to investigate.

The contributions of this paper are as follows: (i) We propose a novel variant of the multiple traveling salesman problem involving communication constraints. (ii) We introduce a mixed-integer convex program (MICP) formulation that utilizes the framework of graphs of convex sets to solve the CRATSP exactly. (iii) We show that the proposed formulation can easily accommodate variants such as the TSP with neighborhoods. (iv) We evaluate the performance of our formulation through extensive numerical experiments. (v) We propose a simple heuristic to find feasible solutions and evaluate the tightness of the lower bounds obtained via

the convex relaxation¹ of our formulation.

II. PROBLEM STATEMENT

Let D denote the starting location, referred to as the depot. This starting location is the same for both the sensing agent A_s and the relay agent A_r . Let $T := \{t_1, \dots, t_n\}$ be the set of targets to be visited by the sensing agent. The locations of the targets and the depot are defined in $\mathbb{R}^2$, and we use t_i and D to represent their position coordinates as well. The communication radius between the agents A_s and A_r , and between the agent A_r and the depot, is denoted by R_{com} . In general, these two communication radii can be different, and the formulations presented here can be trivially extended to accommodate varying radii. The objective of the Communication Relay-Assisted TSP (CRATSP) is to determine the sequence of targets in T to be visited by agent A_s , as well as the locations of the agent A_r corresponding to each target visit by A_s , such that the following conditions are satisfied:

- The sequence of targets begins at the depot D , visits every target in T exactly once, and returns to D .
- When A_s visits a target t_i , the Euclidean distance between t_i and the corresponding location of the relay agent is less than or equal to R_{com} .
- The Euclidean distance between the location of the relay agent and the depot is less than or equal to R_{com} .
- The combined travel cost of the two agents is minimized.

A feasible solution for an instance with four targets is illustrated in Fig. 2. Because we assume that the communication radius is constant in every direction, the relay agent A_r is effectively constrained to lie within a circle of radius R_{com} centered at D . Similarly, when A_s visits a target t_i , A_r is constrained to lie within a circle of radius R_{com} centered at t_i . Therefore, the feasible region for A_r at target t_i is the intersection of these two circles, represented as the almond shapes in Fig. 2. The continuous decision variables that dictate the positions of the relay agent are restricted to these intersections, which are convex sets.

III. MICP FORMULATION

The CRATSP is formulated using the set of vertices $V := T \cup \{D\}$, that includes the targets and the depot. Let $E := \{(i, j), i \neq j, \forall i, j \in V\}$ represent the set of edges. The binary decision variables x_{ij} , $i, j \in V$ determines the sequence of the agent A_s . The variable x_{ij} is equal to 1 if the edge (i, j) is used in the solution, and zero otherwise. The variables u_i , $i \in T$ are used to define sub-tour elimination constraints. The position coordinates of the relay agent A_r when A_s visits target t_i is determined by the variables $z_i \in \mathbb{R}^2$, $i \in T$. The travel cost of an edge (i, j) by the agent A_s is denoted by c_{ij} . Let l_{ij} , $i, j \in V$ be the set of variables denoting the cost of travel for the agent A_r . When agent A_s travels on edge (i, j) , the relay agent A_r travels

¹The convex relaxation of an MICP is a convex optimization problem obtained by removing the integer constraints. The optimal solution of this relaxed problem is a lower bound to the original MICP.

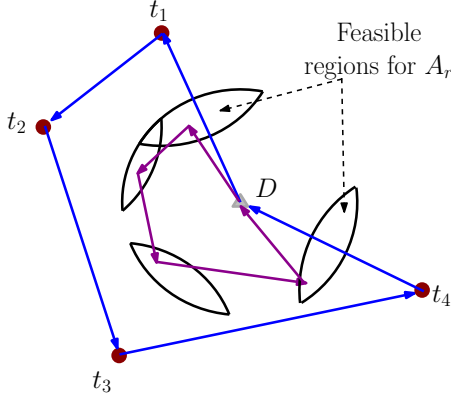


Fig. 2. Example Solution of CRATSP

from z_i to z_j , and the corresponding distance traveled l_{ij} must be equal to $\|z_i - z_j\|$.

A. Baseline formulation for CRATSP

The Mixed-Integer biconvex formulation is given as follows:

$$\mathcal{P}_1 : \text{Minimize } \sum_{i,j \in V} c_{ij} x_{ij} + \sum_{i,j \in V} l_{ij} x_{ij}, \quad (1)$$

subject to:

$$\sum_{j \in V} x_{ij} = 1, \quad \forall i \in V, \quad (2)$$

$$\sum_{j \in V} x_{ji} = 1, \quad \forall i \in V, \quad (3)$$

$$u_i + 1 - M(1 - x_{ij}) \leq u_j, \quad \forall i, j \in T, \quad (4)$$

$$\|z_i - D\| \leq R_{com}, \quad \forall i \in T, \quad (5)$$

$$\|z_i - t_i\| \leq R_{com}, \quad \forall i \in T, \quad (6)$$

$$\|z_i - z_j\| \leq l_{ij}, \quad \forall i, j \in V, \quad (7)$$

$$x_{ij} \in \{0, 1\}. \quad (8)$$

The objective (1) minimizes the total travel cost of the two agents. The two grouped terms correspond to the travel costs of the two agents. The product of the continuous variables l_{ij} and the binary decision variables x_{ij} makes the objective function bilinear. Note that if an edge (i, j) is not used in the solution (i.e., $x_{ij} = 0$), the cost corresponding to both agents identically goes to zero.

The degree constraints, which ensure that each target is entered and exited exactly once, are defined in equations (2) and (3). A complete formulation of the TSP needs to eliminate sub-tours². Without Sub-tour Elimination Constraints (SECs), an optimal solution might yield multiple disconnected loops instead of a single Hamiltonian cycle. The Miller-Tucker-Zemlin (MTZ) constraints shown in (4) [22], [23] eliminate such sub-tours. In (4), M is as a sufficiently large constant, generally referred to as the *big-M*.

²A sub-tour is an invalid disjoint cycle that visits a proper subset of cities and returns to the starting node, without visiting all the cities in the graph.

For each target, the continuous variable z_i represents the position coordinates of the relay agent when that target is visited. To ensure feasible communication, this position must lie within the communication distance from both the depot and the target. Specifically, equations (5) and (6) enforce these communication constraints, formulated as convex second-order cone constraints. Similarly, the convex inequalities in (7) define the length variables l_{ij} for $(i, j) \in E$. By multiplying these lengths by binary variables, the term $l_{ij} x_{ij}$ in the cost function minimizes the distance traveled by the relay agent, i.e., $\|z_i - z_j\|$ when x_{ij} evaluates to 1.

Although there is only one set of variables (z_i) defining the position of A_r at each target, coupling these with the linear distance variables l_{ij} in (7) makes it hard to solve the problem efficiently. Ultimately, the presence of these bilinear terms in the objective and the convex constraints renders this naive formulation computationally inefficient. Therefore, we reformulate the problem entirely using the Graph of Convex Sets (GCS) framework.

B. GCS based MICP formulation

The Graph of Convex Sets (GCS) framework was recently introduced in [24]–[26] as an abstract graph representation wherein the nodes correspond to convex sets. This representation efficiently formulates combinatorial problems coupled with discrete and continuous variables as Mixed-Integer Convex Programs (MICP). While originally developed to address shortest-path problem with obstacles, it later expanded to tackle other combinatorial optimization problems, including the Traveling Salesman Problem (TSP). For the shortest path problem, the decision variables involve determining the path on the graph and the positions inside each convex set that correspond to the nodes on the path. The cost of an edge between two convex sets is a convex function of the variables representing the position within each set. The GCS framework is effectively used to formulate the moving target traveling salesman problem in [27], [28], and is used as a low level planner in [29].

In this section, we develop a GCS-based formulation for the CRATSP. To this end, we introduce the variables $p_{ij}, q_{ij} \in \mathbb{R}^2$ for each edge $(i, j) \in E$. For a given edge (i, j) , p_{ij} and q_{ij} determine the spatial coordinates of relay agent A_r when agent A_s is at the start and end of the edge (i, j) , respectively. In other words, p_{ij} and q_{ij} represent the position of A_r when A_s is at t_i and t_j , respectively. One can interpret the values of p_{ij} and q_{ij} as $z_i x_{ij}$ and $z_j x_{ij}$. Note that these variables represent the corresponding positions only if edge (i, j) is used in the solution; otherwise, they are equal to $(0, 0)$.

In the earlier formulation, the set of variables z_i 's representing the position of A_r must satisfy the communication constraints (5) and (6). These essentially define a convex set of feasible positions for each target, $X_i := \{p \in \mathbb{R}^2 : \|p - D\| \leq R_{com}, \|p - t_i\| \leq R_{com}\}$. If an edge (i, j) is used in the solution, the variables p_{ij} and q_{ij} must be constrained to lie in the convex sets X_i and X_j , respectively. While formulating these constraints typically rely on *big-M* method,

the GCS framework formulates them efficiently without requiring it. They are formulated using the perspective of the convex sets, $\bar{X}_i$'s, as shown below:

$$(p_{ij}, x_{ij}) \in \bar{X}_i, (q_{ij}, x_{ij}) \in \bar{X}_j, \forall (i, j) \in E \quad (9)$$

This is a compact way of representing the following set of constraints for all $(i, j) \in E$:

$$\|p_{ij} - Dx_{ij}\| \leq R_{com}x_{ij}, \|p_{ij} - t_i x_{ij}\| \leq R_{com}x_{ij}, \quad (10)$$

$$\|q_{ij} - Dx_{ij}\| \leq R_{com}x_{ij}, \|q_{ij} - t_j x_{ij}\| \leq R_{com}x_{ij}. \quad (11)$$

In the biconvex formulation, there is only one pair of variables determining the position coordinates of A_r at each target. In the GCS based formulation, at each target t_i , there are multiple variables, $p_{ij}, j \in V \setminus i$, representing the position of A_r . Similarly, the variables $q_{ki}, k \in V \setminus i$ also represents the same position. The constraints in (10), (11) ensure that variables p_{ij} and q_{ij} are non-zero only if the binary variable x_{ij} equals one. This results in two non-zero variables, one from each of the sets $\{p_{ij}, i \in V \setminus \{i\}\}$ and $\{q_{ji}, i \in V \setminus \{i\}\}$. These must be constrained to be equal, similar to the flow conservation constraints. The following node-position conservation constraints ensure this continuity:

$$\sum_{i \in V} q_{ij} - \sum_{i \in V} p_{ji} = 0, \forall j \in T. \quad (12)$$

To formulate the length-defining constraints, we multiply the equation (7) by x_{ij} to obtain $\|p_{ij} - q_{ij}\| \leq l_{ij}x_{ij}$. When x_{ij} is equal to one, this constraint and the objective function together ensure the length of the edge is accounted for in the cost. When x_{ij} is zero, this constraint forces the edge length to be zero. Recall that the convex constraints (10)-(11) also force the variables p_{ij} and q_{ij} to be $(0, 0)$. Therefore, one can drop x_{ij} from this inequality and still obtain the same constraints, as shown below.

$$\|p_{ij} - q_{ij}\| \leq l_{ij}, \forall i, j \in V. \quad (13)$$

The GCS-based mixed-integer convex formulation for the CRATSP can be stated as follows:

$$\mathcal{P}_2 : \text{Minimize: } \sum_{i,j \in V} (c_{ij}x_{ij} + l_{ij}), \quad (14)$$

Subject to:

- Degree constraints (2)-(3),
- SEC constraints (4),
- Comm. relay constraints (9),
- Position conservation constraints (12),
- Length constraints (13).

Lemma 1. *The Mixed integer convex formulations $\mathcal{P}_1$ and $\mathcal{P}_2$ are equivalent.*

Proof. The two formulations share the degree constraints (2)-(3) and the sub-tour elimination constraints (4). These ensure that a valid Hamiltonian cycle is formed by the binary variables x_{ij} .

Formulation $\mathcal{P}_2$ introduces two sets of variables p_{ij} and q_{ij} , which define the position of agent A_r at the start and

end of the edge (i, j) , respectively. The constraints in (9) formulate the communication distance constraints on the p_{ij} and q_{ij} variables. These constraints also force the position variables to zero if edge (i, j) is not chosen, ensuring that only the variables corresponding to the edges in the solution have non-zero values.

The position at a node j is represented by two different variables, one non-zero p_{ij} and one non-zero q_{jk} . The node position conservation constraints (12) ensure consistency, and effectively define a single position for A_r when A_s visits node t_j .

For the edges that are not chosen, the LHS of the length defining constraints (13) becomes zero. Since l_{ij} variables must be non-negative and the objective function minimizes the total length, the optimal solution will set the corresponding l_{ij} variables to zero. Therefore, formulation $\mathcal{P}_2$ exactly models the CRATSP. $\square$

Lemma 2. *The optimal path is recoverable from the solution of the MICP $\mathcal{P}_2$.*

Proof. The tour of the sensing agent A_s is identified by the non-zero values of the binary variables x_{ij} . Since p_{ij} is zero for all unused edges, the position of the relay agent at target t_i is recovered by the sum $\sum_{j \in V} p_{ij}$. $\square$

Remark 1. *The CRATSP formulation can be extended to the TSP with convex neighborhoods (TSPN) with a few modifications. In the TSPN, each city is defined as a convex set in $\mathbb{R}^2$. The agent must visit one point in each city before returning to the start location. In this setup, the variables x_{ij} and the corresponding degree and sub-tour elimination constraints represent the agent's tour. The communication relay constraints are replaced by constraints on p_{ij}, q_{ij} , that restrict these variables to the convex set corresponding to each city. The objective is the tour-length, formulated as $\sum_{i,j \in V} l_{ij}$.*

C. Generalization to Three-Agent Problem

The formulation for two agents in the Section III-B can be easily generalized to multiple agents. We state this for the three-agent setup, where two relay agents A_{r_1} and A_{r_2} are required for the communication. The communication links are established via three hops, depot to A_{r_1} , A_{r_1} to A_{r_2} , and finally A_{r_2} to A_s . To this end, we need two additional sets of variables representing the positions of the relay agents. We introduce the variables $r_{ij}, s_{ij}, i, j \in V$ denoting the position of the agent A_{r_2} when agent A_s is at the start and end of edge (i, j) . The variables p_{ij} and $q_{i,j}$ remain the same as in the formulation in Section III-B. We use the variables l_{ij}^1 and l_{ij}^2 to represent the length traveled by the two relay agents. In this new setup, the communication relay constraints between A_{r_1} and the depot are the same as in (10). The constraints between the relay agents, and between

A_{r_2} and A_s , are formulated as follows:

$$\|p_{ij} - r_{ij}\| \leq R x_{ij}, \forall i \in T, j \in V \quad (15)$$

$$\|q_{ij} - s_{ij}\| \leq R x_{ij}, \forall i \in V, j \in T \quad (16)$$

$$\|r_{ij} - t_i x_{ij}\| \leq R x_{ij}, \forall i \in T, j \in V \quad (17)$$

$$\|s_{ij} - t_j x_{ij}\| \leq R x_{ij}, \forall i \in V, j \in T \quad (18)$$

The node position conservation constraints for this setup are given below:

$$\sum_{i \in V} q_{ij} - \sum_{i \in V} p_{ji} = 0, \forall j \in T. \quad (19)$$

$$\sum_{i \in V} s_{ij} - \sum_{i \in V} r_{ji} = 0, \forall j \in T. \quad (20)$$

The constraints defining the length traveled by the two relay agents are formulated using l_{ij}^1 and l_{ij}^2 , similar to those in Section III-B. These are stated as follows:

$$\|p_{ij} - q_{ij}\| \leq l_{ij}^1, \forall i, j \in V, \quad (21)$$

$$\|r_{ij} - s_{ij}\| \leq l_{ij}^2, \forall i, j \in V. \quad (22)$$

This completes the formulation for the three-agent, three-hop routing problem, and is summarized as follows:

$$\text{Minimize: } \sum_{i,j \in V} (c_{ij} x_{ij} + l_{ij}^1 + l_{ij}^2), \quad (23)$$

subject to:

Degree constraints (2)-(3),

SEC constraints (4),

Comm. relay constraints (10), (15)-(18),

Position conservation constraints (19) - (20),

Length constraints (21)-(22).

IV. COMPUTATIONAL EXPERIMENTS

To evaluate the performance of the MICP formulation, we conduct computational experiments on several randomly generated instances. We generate small instances with 10, 15, and 20 targets to find exact solutions. Since the MICP does not scale well for larger instances, we present a simple heuristic to find high-quality feasible solutions for these cases. Additionally, we use the convex relaxation of the MICP to compute tight lower bounds for large instances, which allows us to evaluate the quality of the feasible solutions produced by the heuristic.

A. Heuristic Solution

The main factor that makes this problem computationally difficult is the coupling between the tours of the two agents and the cost being a function of their positions. To generate a feasible solution quickly, we decouple these variables. Specifically, we first generate a tour for the sensing agent A_s using the LKH³ [30] heuristic. Once the tour of the sensing agent A_s is determined, the remainder of the problem reduces to a simple convex program. Let $S := \{s_1, \dots, s_n\}$

be the sequence of targets visited by agent A_s . We define the variables $y_i \in \mathbb{R}^2$ for $i \in \{1, \dots, n\}$ to represent the position of the relay agent A_r when A_s visits target s_i . The following convex program determines the positions of A_r that minimize its total tour length:

$$\text{Minimize: } \sum_{i=1}^{n-1} l_{ii+1} + l_{1n}, \quad (24)$$

subject to:

$$\|y_i - D\| \leq R_{com}, \forall i \in 1 \dots n, \quad (25)$$

$$\|y_i - s_i\| \leq R_{com}, \forall i \in 1 \dots n, \quad (26)$$

$$\|y_i - y_{i+1}\| \leq l_i, \forall i = 1 \dots n-1, \quad (27)$$

$$\|y_n - y_1\| \leq l_{1n}. \quad (28)$$

B. Computational Results

The locations of the targets are generated randomly from a uniform distribution in a 200×200 planning space, with the depot located at $(0, 0)$. The communication radius is a critical parameter in this problem; therefore, the experiments were performed using communication radii from the set $\{80, 90, 100\}$. For each problem size (*i.e.*, number of targets), the formulation is evaluated on fifty random instances. The MICP is implemented in Julia using the JuMP modeling language [31] and solved with Gurobi. All experiments were conducted on a MacBook Pro equipped with an M3 processor and 36 GB of RAM. The time limit for the MICP solver was set to 1800 seconds for each instance. We tested on a representative set of instances with multiple threads in Gurobi, and found out setting to single thread had the best performance; and therefore, we used single thread for all experiments.

The percentage of instances that converged to the optimal solution within the given time limit is listed in Table I, and the computation times for these converged instances are presented as a box-and-whisker plot in Fig. 3. During the MICP solution process, the solver tracks the best lower bound derived from the convex relaxation and the upper bound obtained from feasible solutions. The progression of the percentage gap between these bounds over time is depicted in Figs. 4(a) and 4(b) for instances with a communication radius of 80. While the majority of instances with 15 targets converged within the time limit, the algorithm successfully solved only 12% of the instances with 20 targets. We observe that the communication radius plays a critical role in the problem's solvability: a larger communication radius creates a larger feasible set, making the problem relatively easier to solve. This is clearly demonstrated by both the higher success rate of problems solved and the decreased computation time required to find an exact solution.

Because the MICP formulation does not scale well for larger instances, we apply the proposed heuristic method to generate high-quality feasible solutions, which serve as upper bounds to the optimal solution. For instances with 30, 40, and 50 targets, we execute the heuristic to obtain these upper bounds, while utilizing the convex relaxation of the MICP-GCS to compute the corresponding lower bounds.

³The Lin-Kernighan-Helsgaun (LKH) algorithm is a widely used heuristic that solves TSP instances with up to 1000 cities in seconds.

The percentage gap between the upper and lower bounds measures the quality of the feasible solution, indicating how close the heuristic's cost is to the optimal solution. In general, a large gap could result from either a low-quality feasible solution or a loose lower bound. Thus, generating tight lower bounds is essential to evaluate the feasible solutions.

The optimality gaps for the larger instances, where the number of targets is 30, 40 and 50, are presented as box-and-whisker plot in Fig. 5(a). A similar plot of the percentage gap for the three-agent CRATSP, formulated in Section III-C, is shown in Fig. 6. The median gap is less than 25% for the two-agent CRATSP instances, and less than 40% for the three-agent instances. These results confirm that the generated bounds are of good quality. We observe that while the gap increases with problem size, it does not degrade significantly. Consistent with our earlier observation, instances with smaller communication radii are more challenging to solve, which is reflected in larger gaps between the upper and lower bounds.

The computation time required to generate feasible solutions via the heuristic is less than a second for all instances and is therefore not explicitly shown. The lower bounds are computed from convex relaxations by limiting the number of explored nodes in the Gurobi branch-and-bound tree to one. This leverages Gurobi's built-in cutting planes to construct tighter root-node bounds. The computation times for these lower bounds are illustrated in Fig. 5(b). With the exception of a few outliers, the average computation time remains under 100 seconds.

TABLE I

PERCENTAGE OF INSTANCES SOLVED EXACTLY BY MICP-GCS WITH
TIME LIMIT OF 1800 SECONDS

Comm. Radius	n=10	n=15	n=20
80	100	84	12
90	100	100	46
100	100	98	86

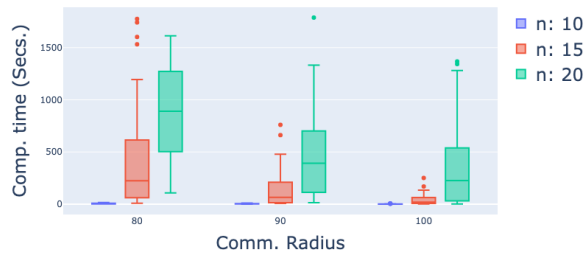
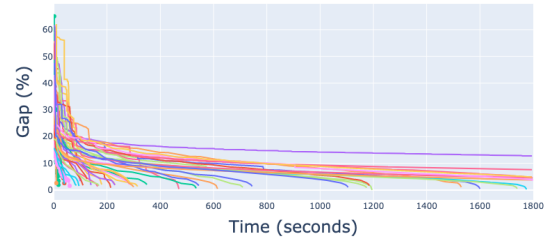


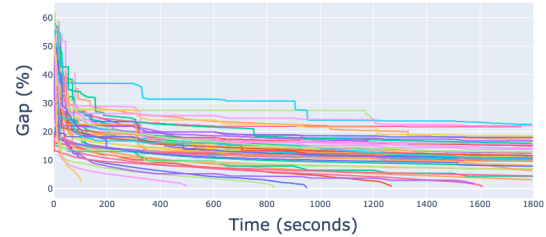
Fig. 3. Computation time required to solve MICP-GCS formulation exactly

V. CONCLUSION

We have presented a cooperative routing problem that arises in surveillance and monitoring applications. The problem involves two agents, where one agent visits a set of targets for data collection, while the second acts as a relay between the sensing agent and the ground station.

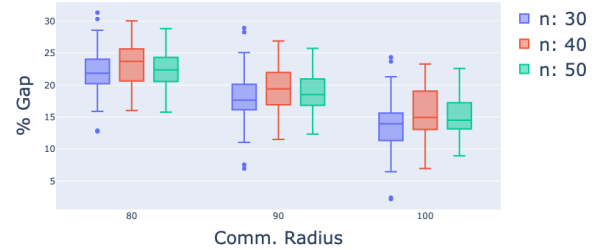


(a) number of targets: 15

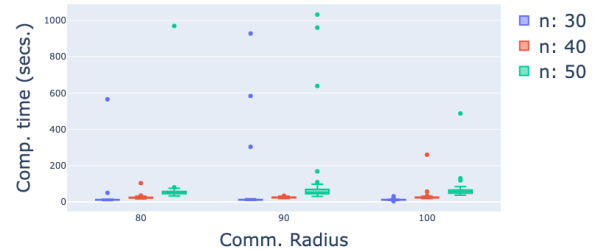


(b) number of targets: 20

Fig. 4. Gap (%) vs. time while solving the MICP-GCS exactly with a time limit of 1800 seconds



(a) Percent gap between upper and lower bound using convex relaxation



(b) Computation time required to solve the root node of the branch and bound tree

Fig. 5. Computation results for convex relaxation of the MICP-GCS formulation

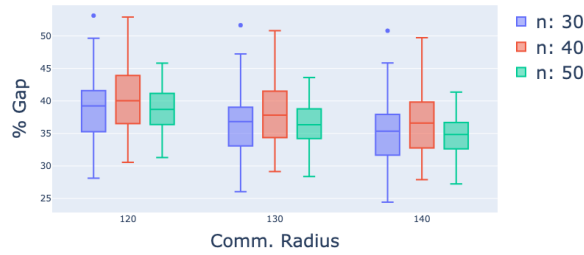


Fig. 6. Percentage gap obtained using the convex relaxation of the CRATSP formulation with three agents

Communication between the agents, as well as between the relay agent and the ground station, is effective only if they are located within a specified threshold distance of each other. The routing problem involves determining the optimal sequence of targets to be visited, along with the continuous locations of the relay agent when each target is visited, in order to minimize the total travel cost of both agents. This problem is formulated as a mixed-integer convex program utilizing the framework of graphs of convex sets, which avoids the use of “big-M” constraints. This approach yields a tight formulation that provides high-quality lower bounds via convex relaxation. For larger problem instances, a simple LKH-based heuristic is proposed to generate feasible solutions. The formulation was tested on several randomly generated instances, and the results of these computational experiments were discussed. The exact MICP formulation is capable of solving instances with up to 20 targets. For larger instances, the median percentage gap between the upper and lower bounds is less than 25%, demonstrating the quality of the feasible solution and the lower bounds.

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